

Homework 4: Spherical Harmonic Analysis

This homework covers material from **Module 4: Spherical Harmonic Analysis** and aligns with learning outcomes **LO4, LO6**.

Related Resources: - Lectures: M4 lecture notes (Spherical Harmonics, Legendre Functions)
- Lab: Lab 4 (Spherical Harmonics) — *Essential for Python implementation of Problems 4.1, 4.2, 4.4* - Textbook: Blakely Ch. 6 - Python libraries: `scipy.special` (Legendre functions)

Problems

1. **Problem 4.1** (Blakely Ch. 6, Problem 1) — *Computational/Easy*

- Compute the first five Legendre polynomials $P_n(\cos \theta)$ for $\theta = 0^\circ, 30^\circ, 60^\circ, 90^\circ$. Plot their behavior.
- **Cross-reference:** Lab 4, Part 1; M4 lecture notes, Section “Legendre Polynomials”
- **Difficulty:** Easy (direct calculation; visualization)
- **Tip:** Use `scipy.special.legendre` or compute from recurrence relation; plot all on same figure

2. **Problem 4.2** (Blakely Ch. 6, Problem 3) — *Derivation/Moderate*

- Show how spherical harmonics are used to represent the gravitational potential of the Earth. Derive the general form.
- **Cross-reference:** Lab 4, Part 2; M4 lecture notes, Section “Application to Laplace’s Equation”; M2/lecture notes (gravity analogy)
- **Difficulty:** Moderate (requires separation of variables; connection to geophysics)
- **Tip:** Start with Laplace’s equation in spherical coords; assume product form $\Phi(r, \theta, \phi) = R(r)\Theta(\theta)\Phi(\phi)$; show Earth potential expansion

3. **Problem 4.3** (Blakely Ch. 6, Problem 5) — *Derivation/Moderate*

- Solve Laplace’s equation in spherical coordinates using separation of variables. Obtain the general solution.
- **Cross-reference:** M4 lecture notes, Section “Separation of Variables”; M1/lecture1_notes.qmd (spherical Laplacian)
- **Difficulty:** Moderate (comprehensive derivation; three ODEs)
- **Tip:** Separate into radial and angular parts; relate angular part to Legendre equation; write general solution as sum over n, m

4. **Problem 4.4** (Blakely Ch. 6, Problem 7) — *Visualization/Moderate*

- Visualize the spherical harmonic $Y_2^1(\theta, \phi)$. Describe its pattern on the sphere.
- **Cross-reference:** Lab 4, Part 3 (visualization code); M4 lecture notes, Section “Surface Harmonics”

- **Difficulty:** Moderate (code/visualization; conceptual description)
- **Tip:** Plot on sphere (e.g., using matplotlib mplot3d or PyVista); describe nodal lines and lobe patterns

5. **Problem 4.5** — *Conceptual/Challenging*

- Explain how spherical harmonic coefficients relate to the spectral content of geophysical fields. What do low-degree vs high-degree harmonics represent?
- **Cross-reference:** Lab 4, Part 2; M6 lecture notes (spectral analysis); M4 lecture notes, Section “Application”
- **Difficulty:** Challenging (requires synthesis of theory and applications)
- **Tip:** Relate low- n to long-wavelength (deep sources); high- n to short-wavelength (shallow sources); discuss resolution limits

Submission Guidelines

- Submit solutions as a PDF or Quarto document (or Jupyter notebook)
- Include derivations, calculations, plots showing convergence behavior
- Use proper mathematical notation for harmonics and functional forms
- Include visualization outputs (e.g., plots of Legendre functions, sphere plots)
- Interpret results physically (connection to geophysical sources)
- Expected length/complexity: equivalent to 5–8 pages of narrative + figures